

METHOD OF MOMENTS

- 1- Given n i.i.d. realizations $\overline{X}_1, \dots, \overline{X}_n$ ^{DATA} from the same distribution $f(x|\theta)$ (assume structure known)
- 2- We want to estimate the parameters $\hat{\theta} = g(\overline{X}_1, \dots, \overline{X}_n)$
- 3- Write θ in terms of moments $\mu_k = E(\overline{X}^k)$ then we replace μ_k with sample moments to get $\hat{\theta}(\hat{\mu}_k)$
estimate $\swarrow$ sample moments

Ex: $\overline{X}_i \sim \text{poisson}(\lambda) = f(x|\theta)$
 $\lambda = E(\overline{X}) = \mu_1$
unknown parameters

Replace w/ $\hat{\lambda} = \hat{\mu}_1 = \frac{1}{n} \sum_{i=1}^n \overline{X}_i = \overline{\overline{X}}$
 $\text{poisson}(\hat{\lambda})$ is a fit from data

Ex: $\overline{X}_i \sim \text{normal}(\mu, \sigma^2) = f(x|\theta)$
 $\mu = E(\overline{X}) = \mu_1$
 $\theta = [\mu, \sigma^2]$

$$\sigma^2 = E(\overline{X}^2) - E(\overline{X})^2 = \mu_2 - \mu_1^2$$

$\Rightarrow$ Replace w/ $\hat{\mu}_1, \hat{\mu}_2$

$$\hat{\mu} = \hat{\mu}_1 = \overline{\overline{X}} \quad \text{sample mean}$$

$$\hat{\sigma}^2 = \hat{\mu}_2 - \hat{\mu}_1^2 = \left(\frac{1}{n} \sum_{i=1}^n \overline{X}_i^2 \right) - \overline{\overline{X}}^2$$